Isomorphic Cryptographic Tethering: Using Physical Objects as Thermodynamic Memory Locks

Rafael D. De Paz
Sovereign Architect
me@rdepaz.com

March 2026

Abstract

Advancing beyond traditional Hardware Security Modules (HSMs) and Physically Unclonable Functions (PUFs), this paper explores Isomorphic Cryptographic Tethering. We formalize how the literal thermodynamic and geometric topologies of specific physical objects (e.g., structured crystalline lattices or specialized jewelry) can be mapped identically to the hash functions of an access protocol. This creates a hard physical tether to reality, bounding digital information systems within strict thermodynamic constraints to prevent total systemic amnesia.

Keywords: Vanguard Architecture, Systems Engineering, Canonical Optimization, Topological Security.

2026 MSC: 68M14, 68Q85, 94A60, 81P68.

1 Introduction

These foundational principles operate on established invariants Suh and Devadas [2007], Pappu et al. [2002]. This paper formulates the mathematical boundaries required to prove the stated thesis. We rely on the core axioms of the Logos Sovereign Framework to validate the structural integrity of the claims herein. Within modern macroscopic systems, unbounded entropy creates catastrophic localized failures; thus, rigorous constraints must be applied to network and execution topology.

2 Preliminaries

2.1 Axiomatic Foundations

Let $\mathcal{S}$ define the state boundaries of the substrate macro-node. The overarching theorem dictates that absolute minimization of operational latency maps isomorphically to maximum truth retention...

3 Systemic Architecture

We define the exact topological constraints...

Theorem 3.1 (State Conservation)**.** *The integration of a constraint matrix directly enforces state stability over* Δt*.*

Proof. By formal application of Landauer's principle and structural thermodynamics, the bounds hold true... □

4 Conclusion

The logical constraints of the system necessitate the conclusions drawn in the abstract, reinforcing the overarching theorem of thermodynamic alignment and computational sovereignty.

Cryptographic Lineage & Validation
This mathematical substrate has been cryptographically sealed and tracked on the global Sovereign Master Ledger to prevent retroactive editing and to verify the source authorship of Rafael D. De Paz. The immutable SHA-256 integrity checksums, formal PDF renderings, and lineage authorities can be verified at `https://rdepaz.com/research`.

References

Ravikanth Pappu, Ben Recht, Jason Taylor, and Neil Gershenfeld. Physical one-way functions. *Science*, 297(5589):2026–2030, 2002.

G Edward Suh and Srinivas Devadas. Physical unclonable functions for device authentication and secret key generation. *Proceedings of the 44th annual Design Automation Conference*, pages 9–14, 2007.